

IbukaPay

Personal Financial Obligation and Bill Tracking System

Project Phase 1 – Project Documentation

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1. Abstract

IbukaPay is a personal financial obligation and bill management system designed to help an individual keep track of money they owe, money owed to them by others, and personal or household bills that need to be paid, all from a single centralized application. Unlike a conventional expense-splitting tool, IbukaPay does not require every financial record to be tied to another registered user. A person, referred to as a Contact, can be linked to a financial record purely as descriptive information, without needing an IbukaPay account of their own. This makes the system realistic for everyday situations such as informal loans, IOUs, and recurring bills like electricity, water, rent, or school fees. The system tracks due dates, payment status, partial payments, reminders, and overdue records, and presents a consolidated dashboard so that the user can understand their overall financial position at a glance.

2. Problem Statement

Many people manage informal financial obligations – money borrowed from or lent to friends and family, and recurring household bills – using memory, scattered notes, or generic chat messages. This approach is unreliable: due dates are forgotten, partial payments are not tracked accurately, and there is no single place to see how much money is owed, how much is expected back, and which bills are outstanding. Existing expense-tracking tools are often built around splitting shared costs between registered members of a group, which forces the user to create an account for every person involved, even when that person is not a real user of the platform (for example, a landlord, a utility provider, or a relative who will never install the app). This creates unnecessary friction and does not reflect how obligations actually work in daily life. There is a need for a lightweight, personal system that lets a single user record any financial obligation – whether it involves another person or not – without forcing that other party to be a registered account holder.

3. Scope of the Project

The project is developed as an academic prototype. The scope defines what will be covered inside the system and what is intentionally left out for this phase.

In scope:

- User registration, login, logout and session protection.
- Recording of financial records under three categories: money the user owes, money owed to the user, and personal/household bills.
- Optional association of a financial record with a Contact (a person or provider who does not need an IbukaPay account).
- Tracking of due dates, payment status (Pending, Paid, Partially Paid, Overdue) and remaining amounts.
- Recording of individual payments made against a financial record.
- Simulated reminders for upcoming and overdue obligations.
- Categorization of records (e.g. Housing, Utilities, Education, Transport).
- Association of bills with a Bill Provider (e.g. an electricity or water company).
- A dashboard summarizing amounts owed, amounts owed to the user, bills due, and overdue totals.
- Basic search and filtering of financial records by type, status and date.
- Validation of amounts, dates and required fields to preserve data integrity.

Out of scope (future work):

- Real SMS/email notifications (reminders remain simulated in this prototype).
- Monthly budgeting and spending statistics.
- Automatic generation of recurring payment records.
- Export of records to external files.
- Support for multiple currencies.

4. AS-IS Model

In the current, informal way of managing personal finances (before IbukaPay), a person typically relies on memory, paper notes, or free-text messages to keep track of obligations. There is no structured record, no automatic reminder, and no consolidated view of the person's financial position.

AS-IS process (informal / manual):

- A person lends or borrows money and mentally notes the amount and the other party.
- Bills (electricity, water, rent, etc.) are remembered from physical receipts, SMS alerts from the provider, or verbal reminders.
- Tracking of who has paid, who still owes, and how much remains is done manually, often inaccurately.
- Due dates are easily forgotten, leading to late payments and strained relationships.
- There is no single place to see total amounts owed versus amounts expected to be received.
- Partial payments are difficult to track without a written or digital record, so remaining balances become unclear over time.

Limitations of the AS-IS approach: information is scattered, error-prone, not searchable, provides no reminders, and gives no overall financial picture.

5. TO-BE Model

With IbukaPay, the same activities are handled through a structured, centralized digital system. The user registers an account and can then create financial records under one of three types: money they owe, money owed to them, or a bill. Each record can optionally be linked to a Contact (a person or organisation) that does not need an IbukaPay account. The system tracks due dates and payment status automatically, supports partial payments, generates simulated reminders, and presents everything on a single dashboard.

TO-BE process (digital / structured):

- User registers and logs in securely to access their personal financial records.
- User creates a financial record and selects its type: I Owe, Owed to Me, or Bill.
- User optionally links the record to a Contact or Bill Provider, entering that party's basic details.
- User sets an amount, a due date, and an optional description/category.
- The system tracks the status of the record (Pending, Partially Paid, Paid, Overdue) and updates the remaining amount as payments are recorded.
- The system generates reminders as due dates approach and flags overdue records.
- The dashboard aggregates all records to show total owed, total owed to the user, total bills due, and overdue totals in one view.
- The user can search and filter records by type, status or date to quickly locate a specific obligation.

This removes the dependency on memory and scattered notes, ensures that obligations involving people outside the platform can still be tracked accurately, and gives the user a single, reliable source of truth for their personal finances.

6. User Stories

User stories are grouped into functional areas covering the full IbukaPay domain. Stories marked in Section J are considered future/advanced scope beyond the academic prototype.

A. Account and Authentication

US-01 Register — As a visitor, I want to create an IbukaPay account so that I can securely manage my financial records.

US-02 Login — As a registered user, I want to log in so that only I can access my financial records.

US-03 Logout — As a logged-in user, I want to log out so that my account is protected when I leave the application.

US-04 Session protection — As a user, I want my financial information to be protected from unauthorised users.

US-05 Unique username — As a user, I want my username to be unique so that my account can be identified correctly.

US-06 Secure password — As a user, I want my password to be securely stored so that my credentials are not exposed.

B. Money I Owe

US-07 Record money I owe — As a user, I want to record money that I owe to someone so that I do not forget my obligations.

US-08 Obligation without an account — As a user, I want to record money I owe to a person who does not have an IbukaPay account.

US-09 Reason for owing — As a user, I want to add a description or reason for an amount I owe so that I understand the purpose of the obligation.

US-10 Set due date — As a user, I want to set a due date for money I owe so that I know when I need to pay it.

US-11 Mark debt as paid — As a user, I want to mark money I owe as paid after I settle it.

US-12 Edit an obligation — As a user, I want to edit an obligation when its information changes.

US-13 Delete an obligation — As a user, I want to delete an incorrect or unnecessary obligation.

US-14 View outstanding obligations — As a user, I want to see all money I currently owe so that I know my outstanding obligations.

US-15 View overdue obligations — As a user, I want to see obligations whose due dates have passed so that I can take action.

C. Money Other People Owe Me

US-16 Record money someone owes me — As a user, I want to record money that another person owes me so that I can keep track of money I expect to receive.

US-17 Person without an account — As a user, I want to record money owed by someone who does not have an IbukaPay account.

US-18 Reason for money owed to me — As a user, I want to record why someone owes me money so that I can remember the transaction.

US-19 Expected repayment date — As a user, I want to set the date when I expect to receive the money.

US-20 Mark money as received — As a user, I want to mark an amount as received after the person pays me.

US-21 Partially paid money — As a user, I want to record a partial payment when someone pays only part of what they owe me.

US-22 View money owed to me — As a user, I want to see all money other people owe me so that I know how much money I am expecting.

US-23 View overdue receivables — As a user, I want to see money that people should have paid me but have not yet paid.

D. Household and Personal Bills

US-24 Record a bill — As a user, I want to record a bill that I need to pay so that I can keep track of my upcoming payments.

US-25 Recurring bills — As a user, I want to record recurring bills such as rent, electricity or internet so that I can remember them every month.

US-26 Set bill due date — As a user, I want to set the due date of a bill so that I know when payment is required.

US-27 Record bill provider — As a user, I want to record who or what organization issued the bill so that I know where the payment goes.

US-28 Mark bill as paid — As a user, I want to mark a bill as paid after making the payment.

US-29 Track unpaid bills — As a user, I want to see all bills that I still need to pay.

US-30 Track overdue bills — As a user, I want to see bills that have passed their due date without being paid.

US-31 Edit bills — As a user, I want to edit bill information when the amount, due date or other details change.

US-32 Delete bills — As a user, I want to delete bills that were entered incorrectly or are no longer relevant.

E. Reminders

US-33 Upcoming payment reminder — As a user, I want to receive a reminder before an obligation or bill is due so that I can prepare for payment.

US-34 Overdue reminder — As a user, I want to be notified when an obligation or bill becomes overdue.

US-35 Reminder for money owed to me — As a user, I want a reminder when someone is expected to repay me so that I can follow up if necessary.

US-36 Choose reminder date — As a user, I want to specify when I should be reminded about a payment.

US-37 Disable a reminder — As a user, I want to disable a reminder when I no longer need it.

F. People / Contacts

US-38 Add a person — As a user, I want to save a person's name and contact information so that I can associate financial records with them.

US-39 Associate person with money I owe — As a user, I want to associate an obligation with a person so that I know whom I need to pay.

US-40 Associate person with money owed to me — As a user, I want to associate money owed to me with a person so that I know who should pay me.

US-41 View person's financial records — As a user, I want to see financial records associated with a person so that I can understand our outstanding transactions.

US-42 Edit person — As a user, I want to update a person's information when their details change.

US-43 Delete person — As a user, I want to delete a person when they are no longer needed, provided that doing so does not break existing financial records.

G. Dashboard

US-44 Financial overview — As a user, I want a dashboard showing my financial obligations so that I can understand my situation quickly.

US-45 Upcoming payments — As a user, I want to see payments that are due soon so that I can prepare for them.

US-46 Overdue summary — As a user, I want to see the total value of overdue obligations so that I know what needs immediate attention.

US-47 Paid summary — As a user, I want to see my recently paid obligations so that I can track completed payments.

US-48 Expected money — As a user, I want to see money that I expect to receive so that I know what is coming back to me.

H. Search and Filtering

US-49 Search records — As a user, I want to search my financial records so that I can quickly find a particular record.

US-50 Filter by status — As a user, I want to filter records by Pending, Paid, Overdue and Partially Paid.

US-51 Filter by type — As a user, I want to filter records by Money I Owe, Money Owed to Me and Bills.

US-52 Filter by date — As a user, I want to filter financial records by due date or date range.

US-53 View all records — As a user, I want to view all my financial records in one place so that I have a complete picture of my obligations.

I. Validation and Data Integrity

US-54 Validate amounts — As a user, I want the system to reject invalid or negative amounts so that my financial records remain accurate.

US-55 Validate dates — As a user, I want the system to validate dates so that invalid payment dates cannot be stored.

US-56 Required information — As a user, I want the system to clearly tell me which information is required when creating a record.

US-57 Preserve data integrity — As a user, I want my financial records to remain consistent when people, bills or obligations are edited or deleted.

J. Future / Advanced Stories

US-58 Monthly budget — As a user, I want to set a monthly budget so that I can control my spending. (Future scope)

US-59 Spending statistics — As a user, I want to see statistics about my financial obligations so that I can understand my financial patterns. (Future scope)

US-60 Categories — As a user, I want to categorize bills and obligations so that I can organize them. (Future scope)

US-61 Recurring payments — As a user, I want recurring payment records to be generated automatically so that I do not have to enter the same bill every month. (Future scope)

US-62 Real notifications — As a user, I want to receive reminders through email or SMS so that I do not miss important payments. (Future scope)

US-63 Export records — As a user, I want to export my financial records so that I can keep a copy outside IbukaPay. (Future scope)

US-64 Payment recording — As a user, I want to record the date and amount of a payment so that I have a history of payments.

US-65 Multiple currencies — As a user, I want to record financial obligations in different currencies so that I can manage payments made in different currencies. (Future scope)

7. Business Requirements

IbukaPay is a personal financial obligation and bill management system that allows users to record, organize and monitor money they owe, money owed to them, and personal or household bills. Financial records can optionally be associated with people or organizations, but those parties do not need to have IbukaPay accounts. The system tracks due dates, payment status, reminders and overdue records in one centralized place.

7.1 Functional Business Requirements

- The system shall allow a visitor to register a new account with a unique username and a securely stored password.
- The system shall allow a registered user to log in and log out, and shall protect all financial data behind authentication.
- The system shall allow a user to create a financial record of type I Owe, Owed to Me, or Bill.
- The system shall allow a financial record to optionally reference a Contact or a Bill Provider, without requiring that party to hold an IbukaPay account.
- The system shall allow a user to specify an amount, due date, description/reason and category for a financial record.
- The system shall track the payment status of each record: Pending, Partially Paid, Paid, or Overdue.
- The system shall allow a user to record one or more payments against a financial record and automatically recompute the remaining balance.
- The system shall automatically mark a record as Overdue once its due date has passed without full payment.
- The system shall generate simulated reminders for upcoming and overdue financial records.
- The system shall provide a dashboard summarizing total amounts owed, total amounts owed to the user, total bills due, and overdue totals.
- The system shall allow the user to search and filter records by type, status and date.
- The system shall allow the user to edit and delete financial records, contacts, bills, and bill providers while preserving data integrity.

7.2 Business Rules

- Every financial record must belong to exactly one User; a financial record cannot exist without an owning user.
- A Contact is optional on a financial record and, where present, is owned by exactly one User.
- A Contact does not require its own IbukaPay account; it is stored purely as descriptive information (name, phone, email).
- A financial record may have zero, one, or many associated Payments; each Payment belongs to exactly one financial record.
- A financial record may have zero, one, or many Reminders; each Reminder belongs to exactly one financial record.
- A Category and a BillProvider are optional classifications on a financial record and may be reused across many records.

- The remaining amount on a financial record must never be negative and must be recalculated whenever a payment is added.

8. Software Qualities Applied in the System

Quality	How it is Applied in IbukaPay
Security	User passwords are stored using a strong hashing approach (e.g. PBKDF2) rather than plain text; sessions are protected so that only an authenticated user can access their own financial records.
Usability	The dashboard provides a clear, consolidated summary of the user's financial position; forms clearly indicate required fields and validation errors.
Reliability	Payment status (Pending, Partially Paid, Paid, Overdue) is derived consistently from stored data, and remaining balances are recalculated automatically to avoid inconsistent states.
Data Integrity	Deleting a Contact, Category or BillProvider that is still referenced by existing financial records is prevented or safely handled so that historical records are not corrupted.
Maintainability	The domain model separates concerns cleanly (User, FinancialRecord, Contact, Payment, Reminder, Category, BillProvider), making the codebase easier to extend with future features such as budgets or statistics.
Scalability	The data model supports an unlimited number of financial records, payments and reminders per user, allowing the system to grow with the user's needs.
Performance	Search and filter operations are designed to operate efficiently over a user's own record set, returning results without unnecessary delay.
Portability	The system is built using standard, widely supported web technologies, allowing it to run in different environments with minimal changes.
Testability	Clear validation rules (amount, date, required fields) and well-defined entity relationships make it straightforward to write unit and integration tests for the core domain logic.

9. Initial Class Diagram

The class diagram below shows the initial set of entities involved in IbukaPay and the cardinality of the relationships between them, expressed from the point of view of each entity (min:max notation).

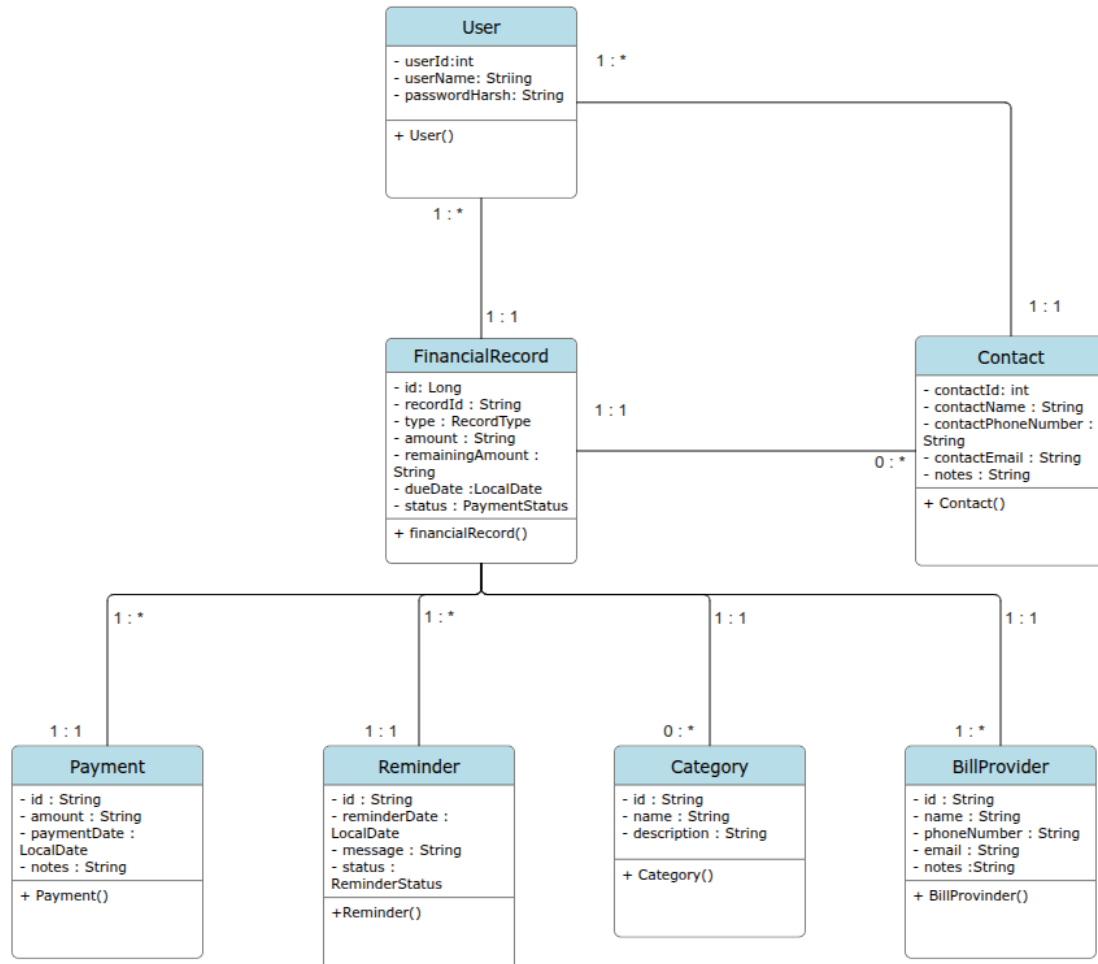


Figure 1. IbukaPay initial class diagram (entities and relationship cardinalities).

9.1 Relationship Cardinalities (per entity)

Relationship	Cardinality	Meaning
User → FinancialRecord	1 : *	One User can have many FinancialRecords
FinancialRecord → User	1 : 1	Each FinancialRecord belongs to exactly one User
User → Contact	1 : *	One User can have many Contacts
Contact → FinancialRecord	0 : *	One Contact can be linked to zero or many FinancialRecords
FinancialRecord → Contact	0 : 1	A FinancialRecord may have no Contact or exactly one Contact
FinancialRecord → Payment	1 : *	One FinancialRecord can have many Payments

Relationship	Cardinality	Meaning
Payment → FinancialRecord	1 : 1	Each Payment belongs to exactly one FinancialRecord
FinancialRecord → Reminder	1 : *	One FinancialRecord can have many Reminders
Reminder → FinancialRecord	1 : 1	Each Reminder belongs to exactly one FinancialRecord
Category → FinancialRecord	0 : *	One Category can be used by many FinancialRecords
FinancialRecord → Category	0 : 1	A FinancialRecord may have no Category or exactly one Category
BillProvider → FinancialRecord	0 : *	One BillProvider can be linked to many FinancialRecords
FinancialRecord → BillProvider	0 : 1	A FinancialRecord may have no BillProvider or exactly one BillProvider

9.2 Entities Overview

Entity	Description
User	The account holder. Owns all financial records and contacts created in the system. Attributes: userId, userName, passwordHash.
FinancialRecord	The central entity representing an obligation. Attributes: id, recordId, type (I Owe / Owed to Me / Bill), amount, remainingAmount, dueDate, status.
Contact	A person or party referenced by a financial record, who does not require an IbukaPay account. Attributes: contactId, contactName, contactPhoneNumber, contactEmail, notes.
Payment	An individual payment made against a financial record. Attributes: id, amount, paymentDate, notes.
Reminder	A simulated notification tied to a financial record. Attributes: id, reminderDate, message, status.
Category	An optional classification for a financial record (e.g. Housing, Utilities). Attributes: id, name, description.
BillProvider	An organisation issuing a bill (e.g. an electricity or water company). Attributes: id, name, phoneNumber, email, notes.

10. Project Links

Demonstration Video:

<https://docs.google.com/videos/d/1QMHXo4d83JaWtFY4iMgtLXWjoD5XvSRJd4avphjVbzw/play?usp=s>
haring

Source Code Repository:

<https://github.com/KimGakuba/lbukaPay>